

Vinh Ly

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Summary

Software Engineer with **18 years** designing and shipping large-scale distributed systems at Google, YouTube, and Whatnot. Owned end-to-end infrastructure serving **2B+ daily video requests** at YouTube, led platform teams scaling real-time systems to **10M+ concurrent users** at Whatnot, and managed a 12-engineer team driving **\$50M in ad revenue** via performance improvements at Google. Built on-call cultures across three organizations and partnered with product, data science, and executive leadership to align engineering with business outcomes. Next focus: senior IC or staff role driving platform reliability and developer experience at a product-led company.

Core Skills

Languages: Python, C++, Java, Go, SQL, TypeScript, JavaScript, HTML, CSS

Infrastructure: Kubernetes, GCP, AWS, distributed systems, microservices

Databases: Bigtable, Spanner, Redis, MySQL, Cassandra

Practices: on-call reliability, incident management, performance optimization, SLO/error budgets, cross-team collaboration

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Designed and owned video serving infrastructure handling **2B+ daily requests**, driving latency reductions via kernel-level tuning and protocol optimization.
- Built automated rollback and canary deployment systems reducing rollout failures by **60%** across 50+ services in a multi-tenant Kubernetes cluster.
- Partnered with data science to instrument caching layers, improving cache hit rates by **25%** and reducing backend load.
- Established on-call rotation and runbook standards for platform team supporting **12 internal product teams**.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Owned migration of monolith to **microservices** on **Kubernetes**, reducing deployment times from **45 minutes to 8 minutes**.
- Led platform infrastructure team scaling real-time auction systems from **100K to 10M+ concurrent users** during rapid growth phase.
- Redesigned event-driven architecture using **Kafka** and **Redis**, cutting p99 latency from **800ms to 120ms** for live bidding.
- Mentored 4 engineers and established performance review calibration process for the platform org.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Led incident command for **50+ SEV1 outages**, reducing mean time to resolution from **45 minutes to 18 minutes** via automation.
- Managed team of **12 engineers** owning ad targeting infrastructure generating **\$50M+ annual revenue** via latency and relevance improvements.
- Partnered with product and data science to define **SLOs and error budgets** adopted org-wide across 200+ services.
- Cut infrastructure costs by **\$4M/year** via node bin-packing optimization and **Spanner** query pattern analysis.
- Designed **Bigtable**-based feature store serving **100M+ requests/second** with sub-millisecond p99 latency requirements.

Software Engineer | Google | 02/2011 - 02/2015

- Developed automated testing framework reducing QA cycle time by **30%** and catching **90% of regressions** pre-deploy.
- Optimized disk I/O patterns in log processing pipeline, improving throughput by **40%** on **10K+ machine** fleet.
- Collaborated with kernel networking team to tune TCP stack for improved throughput in distributed training workloads.
- Built **C++** serving infrastructure for search ranking features used by **1B+ users** globally.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Delivered database migration tooling supporting **MySQL to SQL Server** transitions for **10 enterprise customers**.
- Developed **C++** compiler optimization passes reducing binary size by **15%** for Windows core components.
- Built internal performance profiling tooling adopted by **200+ engineers** across Windows division.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Implemented **C++** unit test framework integration for Visual Studio team, reducing test authoring time by **25%**.

Education

B.S. Honors

Virginia Tech | 2004 - 2007